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Bihar Engineering University, Patna

B.Tech 1st Semester Special Exam., 2020

Subject: Mathematics—I (Calculus and Linear Algebra)

Code: 102102/105102

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct answer of the following (any seven) : **[2×7=14]**

(a) If $Y = \int_0^\infty x^a/a^x \, dx$, $a > 1$, then the value of Y is

(i) $[\Gamma(a)] / [[\log_e a]^a]$
(ii) $[\Gamma(a+1)] / [[\log_e a]^a]$
(iii) $[\Gamma(a+1)] / [[\log_e a]^{a+1}]$
(iv) $[\Gamma(a)] / [[\log_e a]^{a+1}]$

(b) The area bounded by the axis of x, and the curve and ordinates $y = \cosh x/c$ from $x = 0$ to $x = a$ is

(i) $\cosh a/c$
(ii) $\sinh a/c$
(iii) $\sinh a/c$
(iv) None of the above

(c) Consider the following functions :

1. $y = x\sin 1/x$, $x \neq 0$; $and y = 0 if x = 0$
2. $y = x^2\sin 1/x$, $x \neq 0$; $and y = 0 if x = 0$
3. $y = x^2\cos 1/x$, $x \neq 0$; $and y = 0 if x = 0$
4. $y = x\cos 1/x$, $x \neq 0$; $and y = 0 if x = 0$

The functions, differentiable at $x = 0$, are

(i) 1 and 2
(ii) 2 and 3
(iii) 3 and 4
(iv) 1 and 4

(d) For a positive term series $\sum a_n$, the ratio test states that

(i) the series converges, if $\lim_{n\to\infty}[a_{n+1}] / [a_n] > 1$
(ii) the series converges, if $\lim_{n\to\infty}[a_{n+1}] / [a_n] < 1$
(iii) the series converges, if $\lim_{n\to\infty}[a_{n+1}] / [a_n] = 1$
(iv) None of the above

(e) If $\lim_{x\to\infty}[\sin 2x + a\sin x] / [x^3] = b$ where b is finite, then the values of a and b respectively will be

(i) (-2, -1)
(ii) (2, 1)
(iii) (-2, 1)
(iv) (2, -1)

(f) The expansion of $\tan x$ in powers of x by Maclaurin's theorem is valid in the interval

(i) $(-\infty, \infty)$
(ii) $(-3\pi/2, 3\pi/2)$
(iii) $(-\pi, \pi)$
(iv) $(-\pi/2, \pi/2)$

(g) The value of $\lim_{(x, y)\to(k, 0)}(1 + x/y)^y$ is

(i) 1
(ii) e^{-k}
(iii) e^k
(iv) Does not exist

(h) The gradient of the function $f(x, y, z) = \sin(xyz)$, at $(1, -1, \pi)$, is

(i) $\pi(i - j + k)$
(ii) $\pi(i + j + k)$
(iii) $(i + j + k)$
(iv) $(\pi i - \pi j + k)$

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(i) If det(A) = 7, where A = $\begin{vmatrix} a & b & c \\ 1 & 1 & g \\ g & \omega & 1 \end{vmatrix}$, then det(2A)^{-1} is equal to

(i) 14/49 (iii) 7/2

(ii) 1/56

(j) If $3x + 2y + z = 0$, $x + 4y + z = 0$ and $2x + y + 4z = 0$ be a system of equations, then

(i) it is inconsistent

(ii) it has only the trivial solution (0, 0, 0)

(iii) it can be reduced to a single equation and so a solution does not exist

(iv) the determinant of the matrix of coefficients is zero

Q.2 Solve both questions : **[14]**

(a) Evaluate $\int_0^{\infty} \log(x + 1/x)[dx] / [1 + x^2]$

(b) Find the volume of the solid generated by rotating completely about the x-axis where the area enclosed between $y^2 = x^3 + 5x$ and the line $x = 2$ and $x = 4$ about its major axis.

Q.3 Solve both questions : **[14]**

(a) Find the maximum value of the function $f(x) = [x] / [1 + x\tan x]$

(b) It is given that Rolle's theorem holds for the function $f(x) = x^3 + bx^2 + cx$, $1 \le x \le 2$ at the point $x = 4/3$. Find the values of b and c.

Q.4 Solve both questions : **[7+7=14]**

(a) Discuss the convergence of the sequence whose n-th term is $\alpha_n = [(-1)^n] / [n] + 1$

(b) Test the convergence of the following series : $x^2 + 2^2x^4/3.4 + [2^2 4^2 x^6] / [3.4.5.6] + [2^2 4^2 6^2 x^8] / [3.4.5.6.7.8] ...$

Q.5 Solve both questions : **[7+7=14]**

(a) Find the Fourier series expansion of the function $f(x) = \{x^2, -2 \le x \le 2\}$. Hence deduce that $\pi^2/6 = 1/1^2 + 1/2^2 + 1/3^2 + 1/4^2 + ...$

(b) Find the Fourier cosine series and Fourier sine series of the following function in given interval :
$f(x) = x$, & $0 < x < 2$
2, & $2 \le x < 4$

Q.6 Solve both questions : **[7+7=14]**

(a) Discuss continuity of the following function at the point (0, 0) : $f(x, y) = [x^2y^2] / [(x^3 + y^3)]$, & $(x, y) \ne (0, 0)$
0, & $(x, y) = (0, 0)$

(b) Find the maximum value of xyz under the constraints $x^2 + z^2 = 1$ and $y - x = 0$.

Q.7 Solve both questions : **[10+4=14]**

(a) Find the value of $\lim_{x \to \infty} ([x+4] / [x+2])^{x+3}$

(b) Find the equation of the tangent plane to the surface $x^2 - 3y^2 - z^2 = 2$, at the point (3, 1, 2).

Q.8 Solve this question : **[14]**

Find the eigenvalues and eigenvectors of the following matrix : $\begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix}$

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Q.9 Solve both questions : **[7+7=14]**

(a) Verify Cayley-Hamilton theorem for the matrix $\begin{vmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ -2 & 1 & 4 \end{vmatrix}$

(b) Determine the range of the following linear transformation. Also find the rank of T, where it exists.
T: V_2toV_3 defined by $T(x_1, x_2) = (x_1, x_1 + x_2, x_2)$

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